

Compressed data structures for Heegaard splittings

Henrique Ennes and Clément Maria
International Symposium on Computational Geometry
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Compressed data structures for Heegaard splittings (of closed 3-manifolds)

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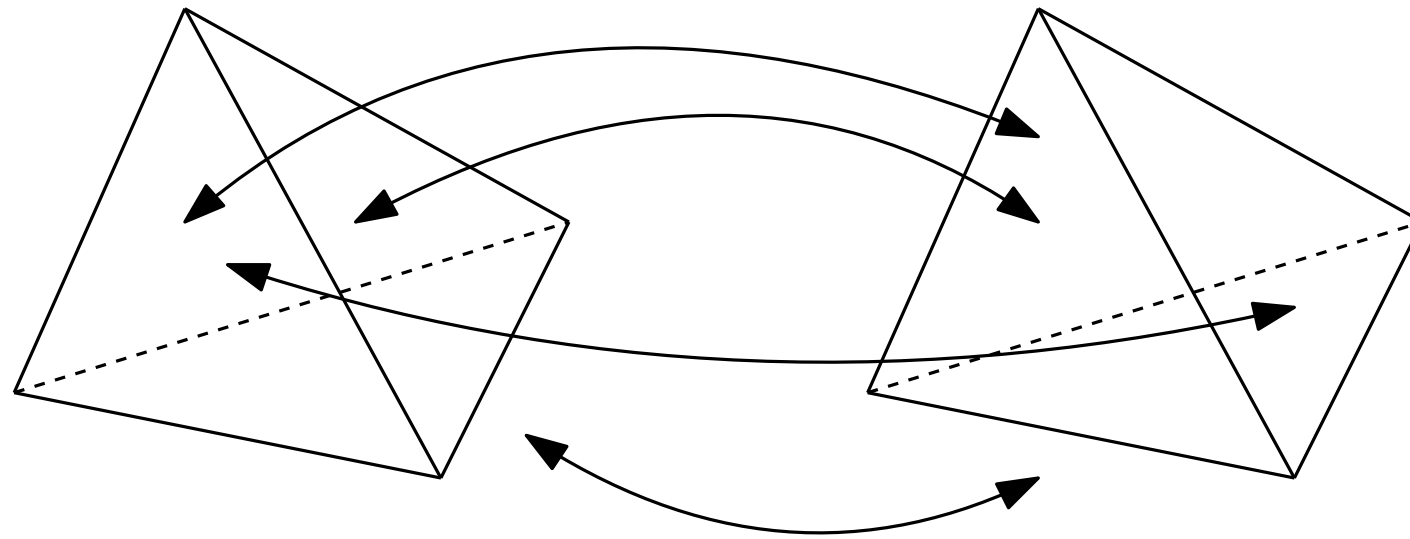


Classifying 3-manifolds

- For constructing a **census** of topological spaces, one needs
 1. **compact data structures** to represent massive amounts of candidate spaces
 2. **efficient operations** to simplify the representation combinatorics
 3. methods for **discriminating spaces**, such as invariants

Classifying 3-manifolds

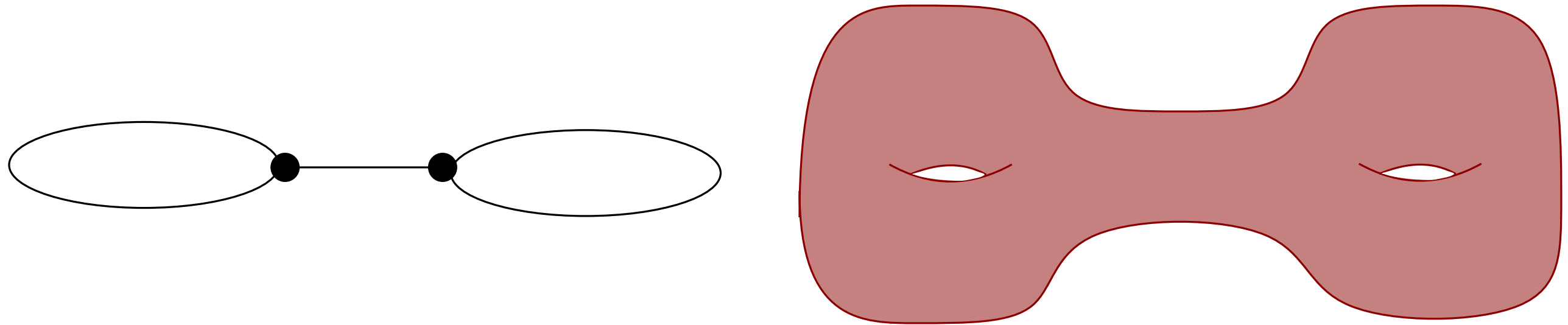
- For constructing a **census** of topological spaces, one needs
 1. **compact data structures** to represent massive amounts of candidate spaces
 2. **efficient operations** to simplify the representation combinatorics
 3. methods for **discriminating spaces**, such as invariants
- The usual data structure is a **triangulation**



Complexity is usually measured on the number of tetrahedra

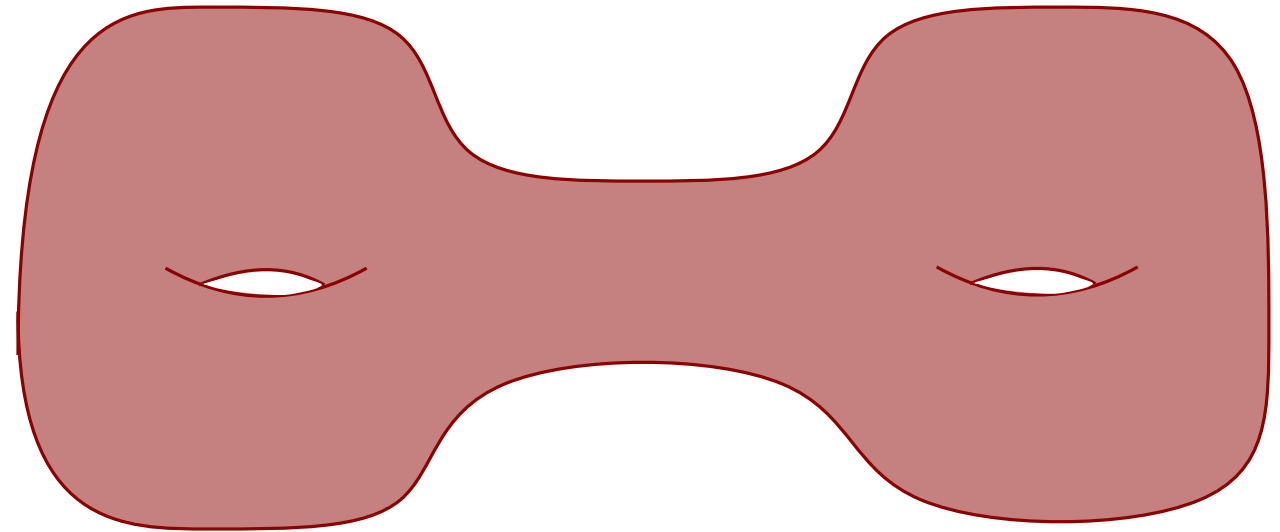
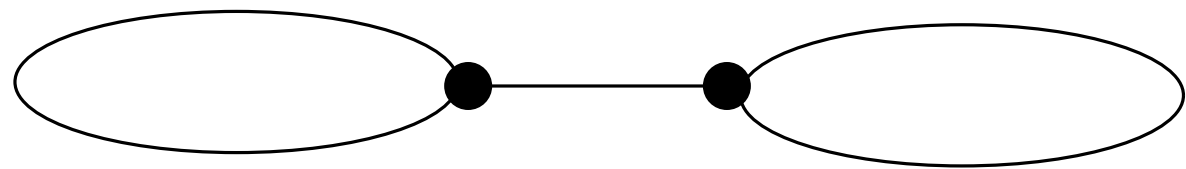
Heegaard splittings

- A **handlebody** is the thickening of a graph in $\mathbb{R}^3$

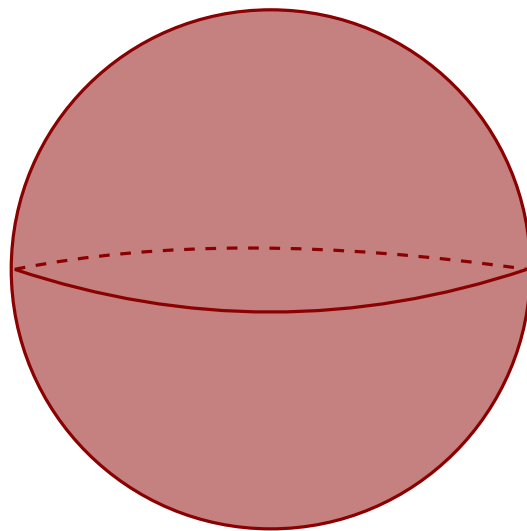


Heegaard splittings

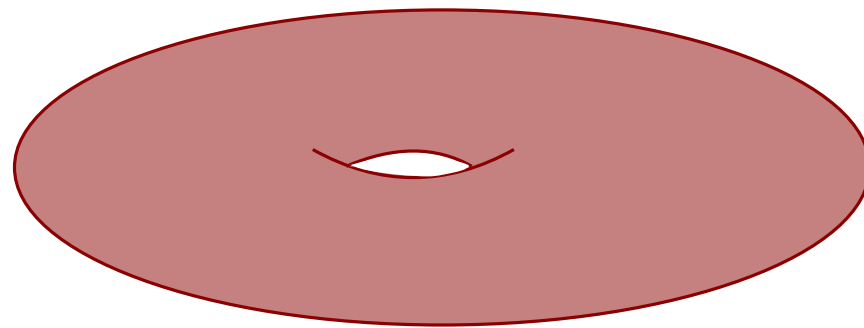
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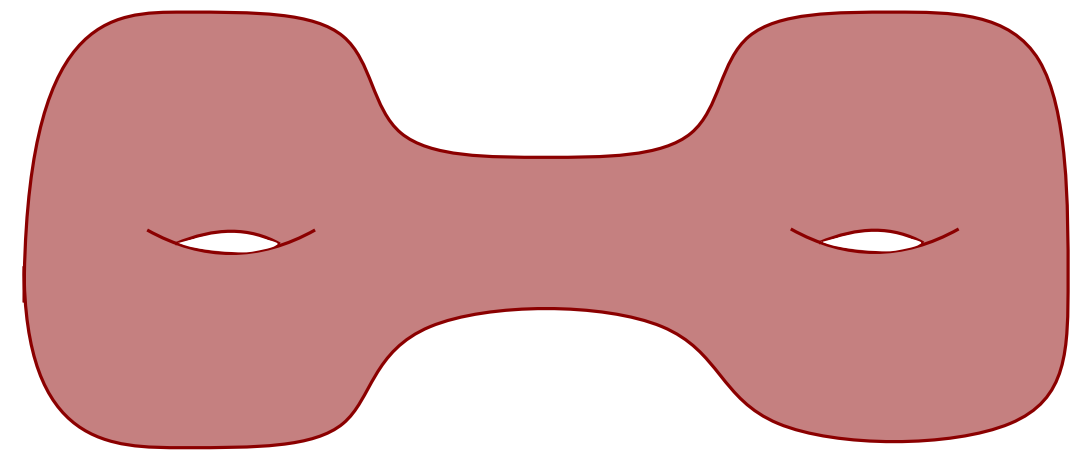
- Handlebodies are uniquely defined by their g **surfaces** as boundaries



$g = 0$



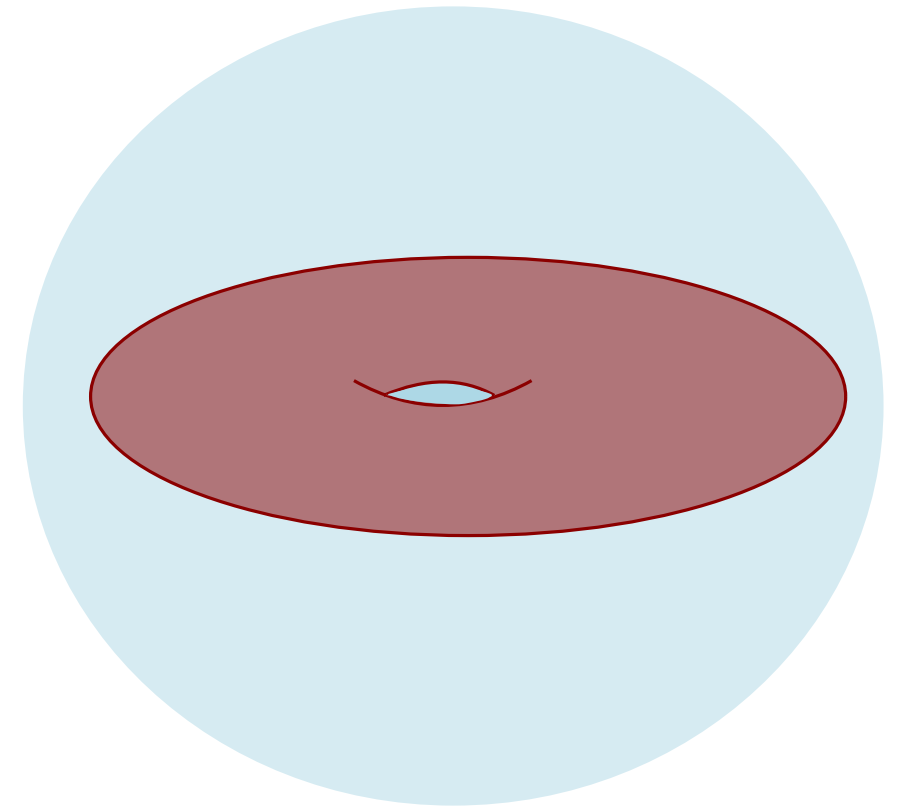
$g = 1$



$g = 2$

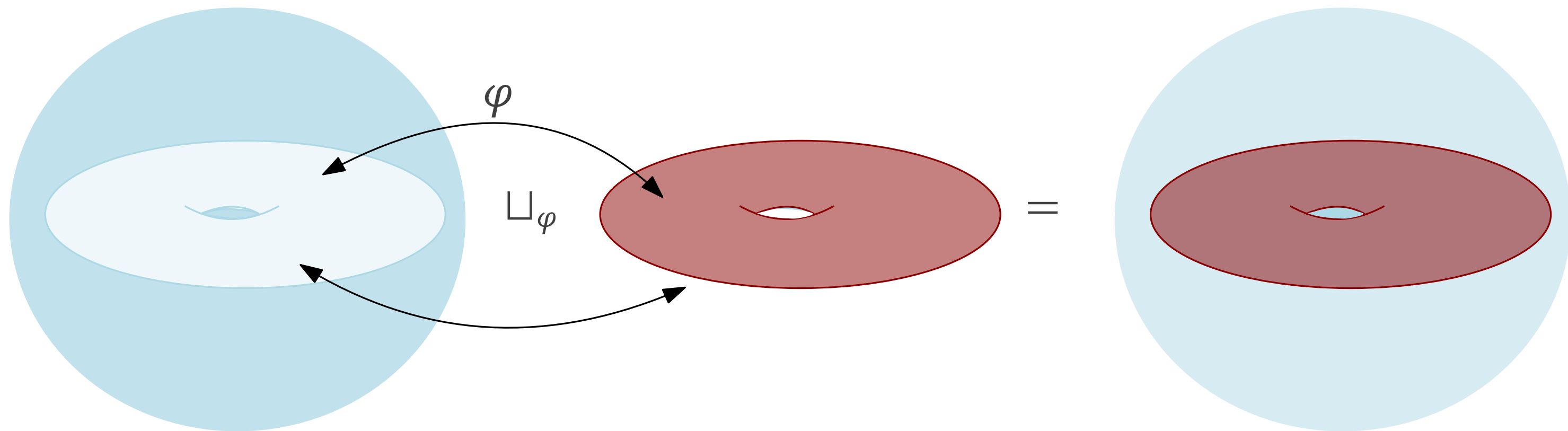
Heegaard splittings

- The sphere S^3 can be seen as the gluing of two handlebodies of genus 1



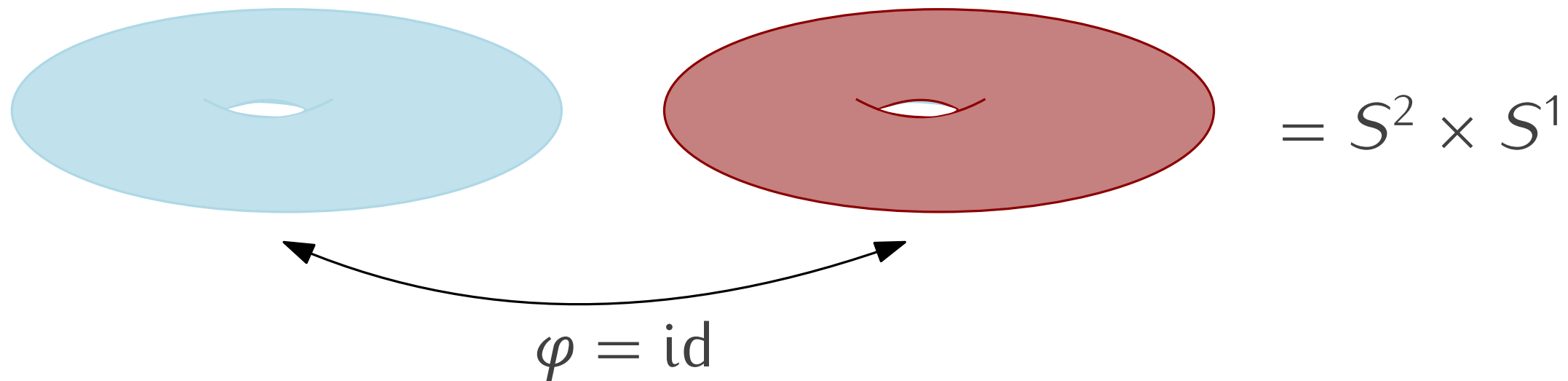
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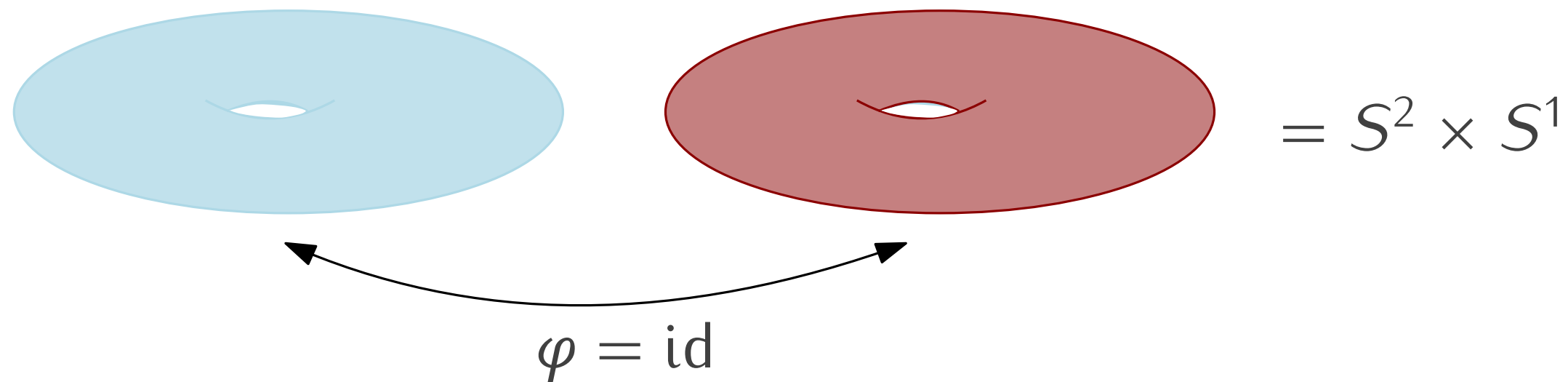
Heegaard splittings

- By changing the gluing map $\Sigma_g \rightarrow \Sigma_g$, we get different constructions of closed 3-manifolds



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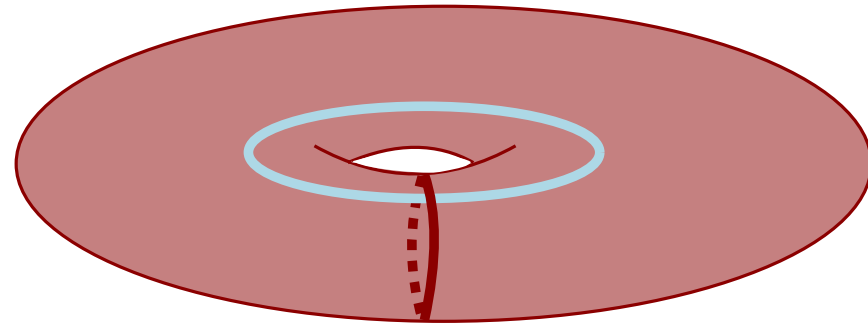


- A **Heegaard splitting** corresponds to the a tuple (Σ_g, φ) with $\varphi \in \text{Mod}(\Sigma_g)$, where we **glue** the handlebodies along the boundaries Σ_g using φ

Every closed 3-manifold has a Heegaard splitting

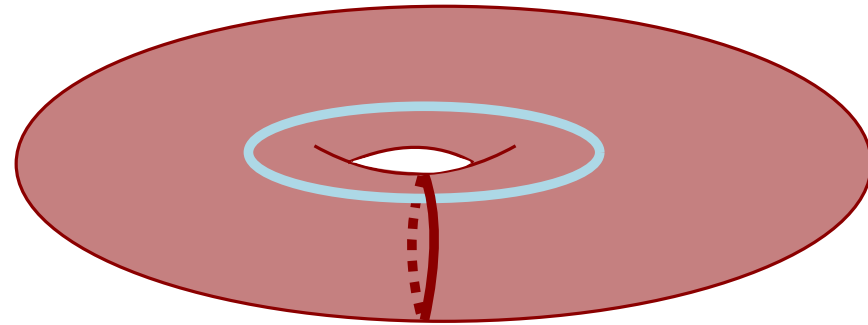
Heegaard splittings

- A **meridian** is a curve that bounds a disk inside the handlebody

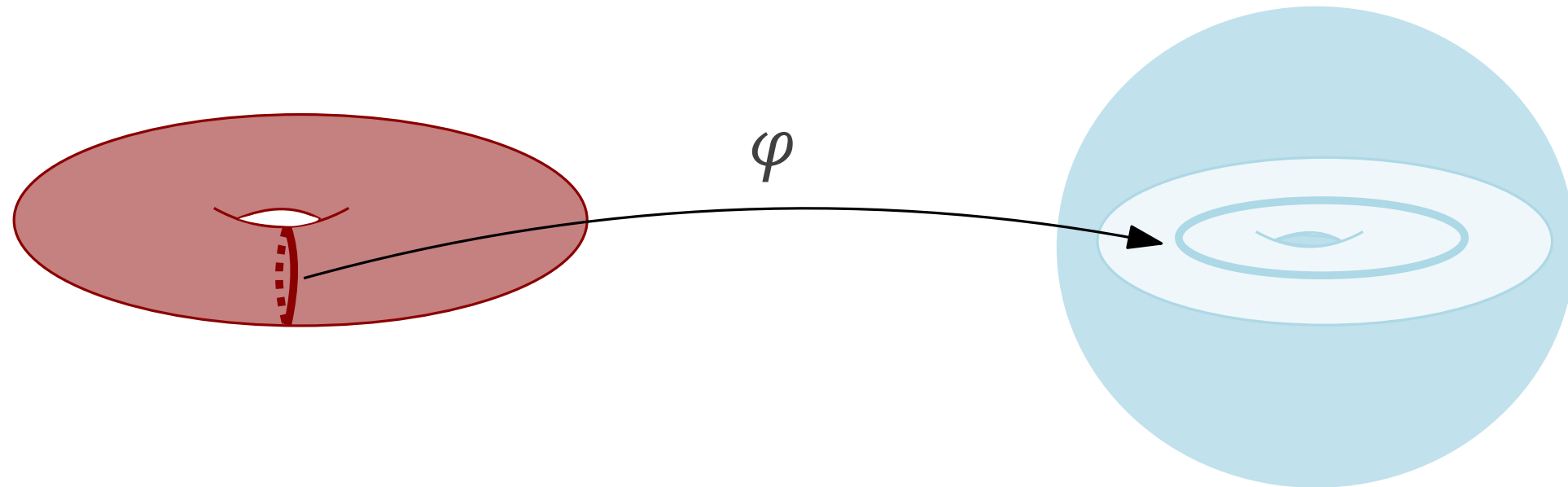


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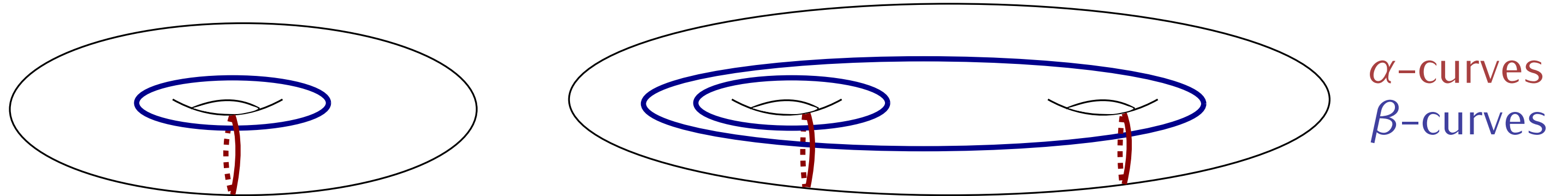


- The gluing map φ takes meridians of one handlebody to the other's



Heegaard splittings

- A **Heegaard diagram** represents a splitting by showing the meridians on the gluing surface

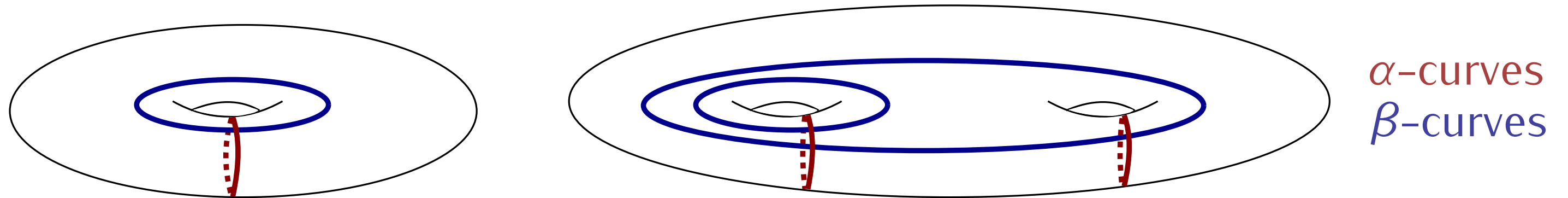


Every closed 3-manifold has a Heegaard diagram

The more α and β intersect, the more complex is the manifold

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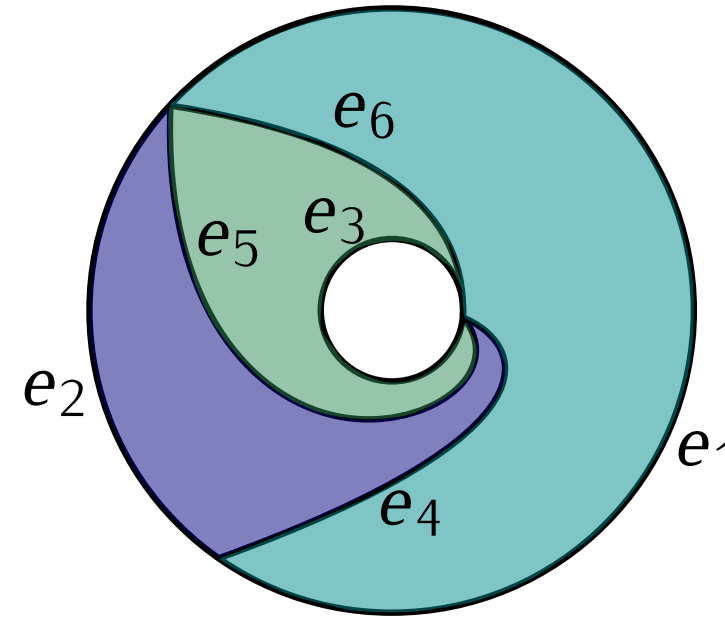
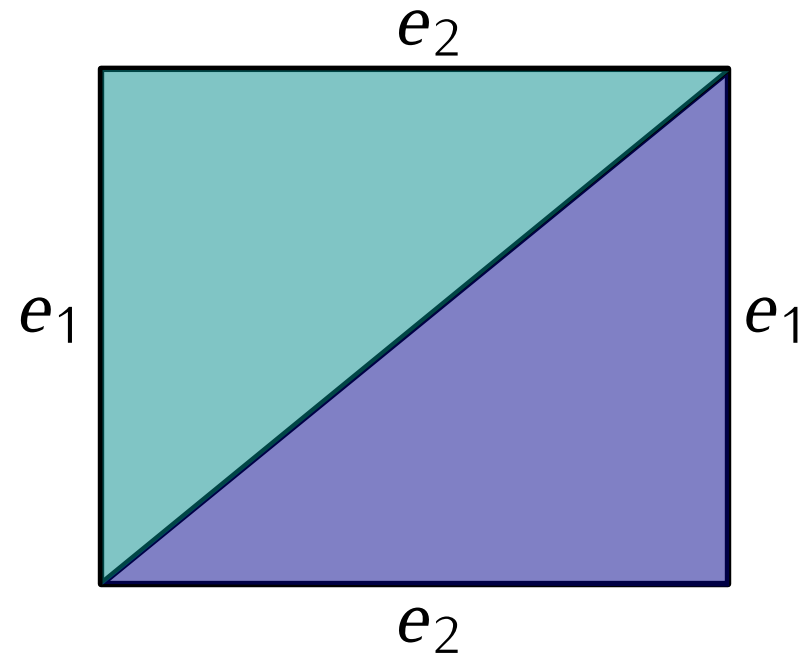
Every closed 3-manifold has a Heegaard diagram

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- A Heegaard diagram can be described by $(\Sigma_g, \alpha, \beta)$

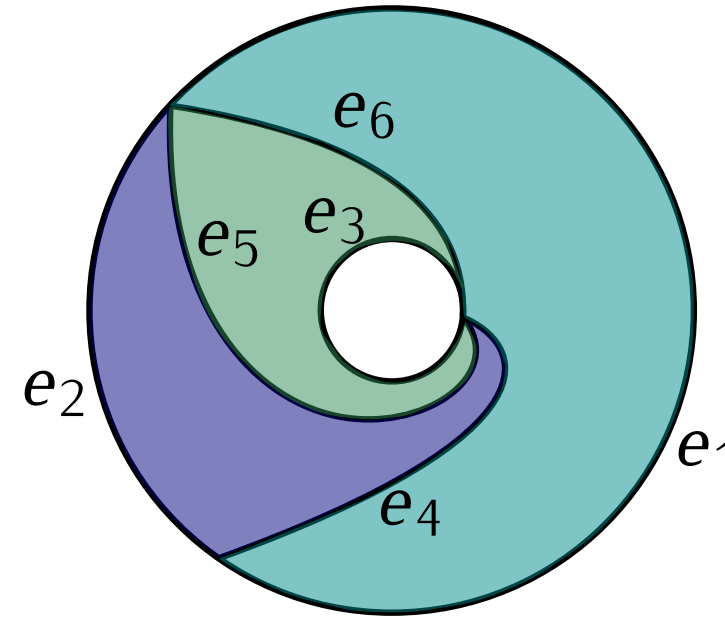
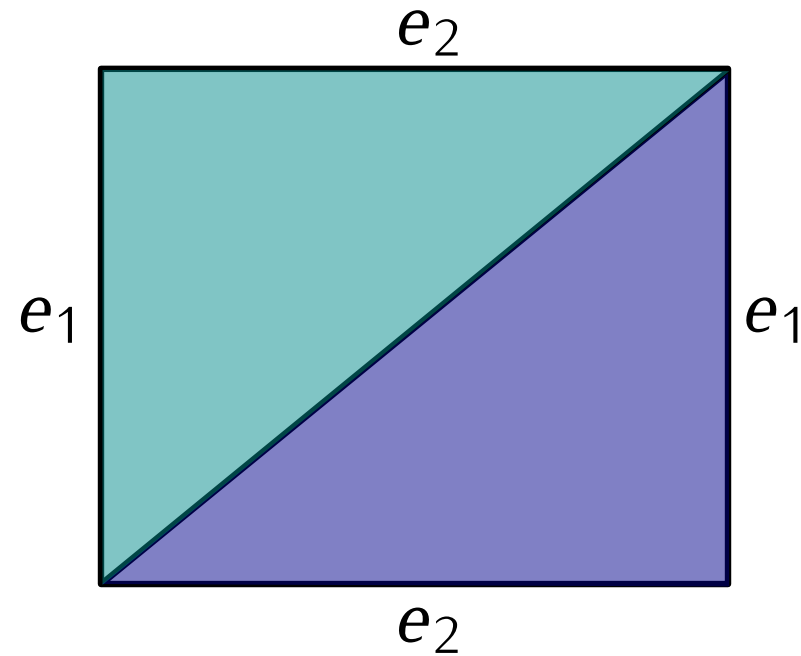
A data structure for Heegaard diagrams

- A **triangulation** of a surface Σ_g is a graph whose faces are bounded by 3 edges each

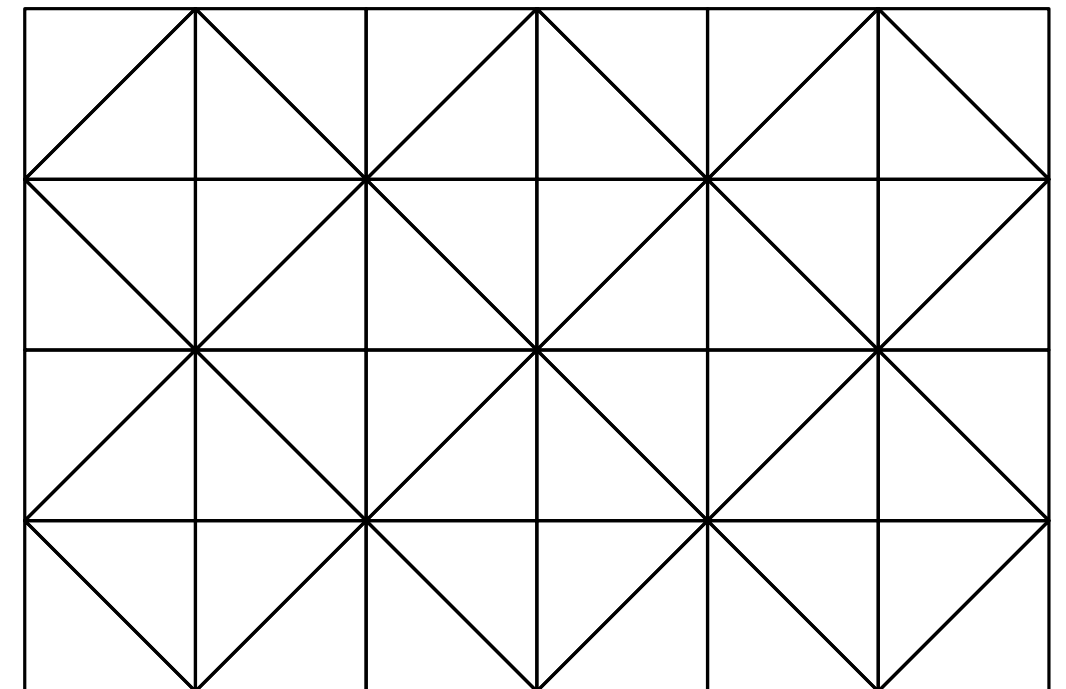


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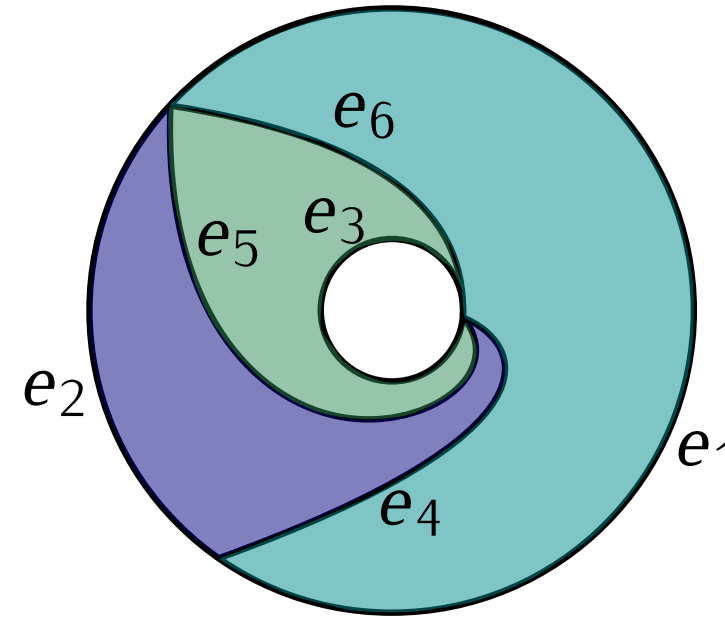
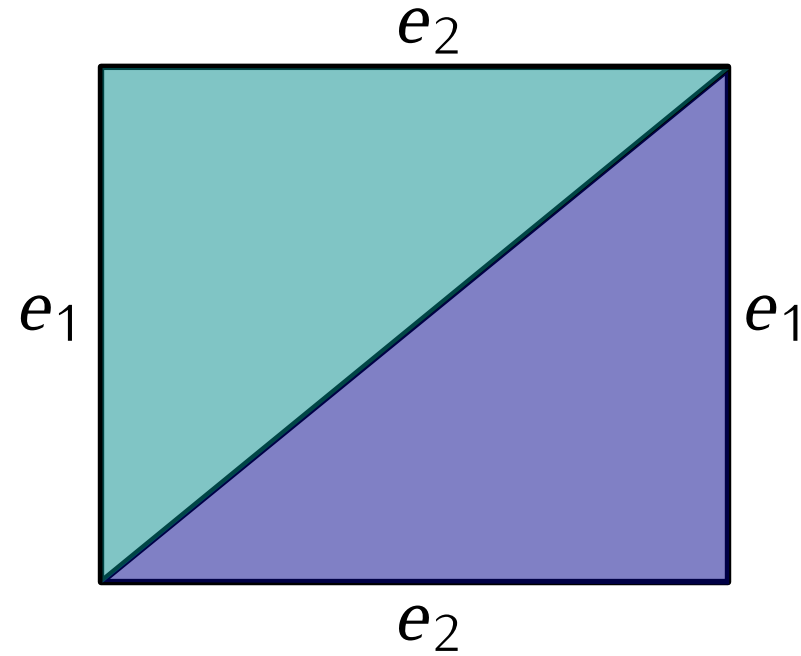


- A curve in a triangulation can be either

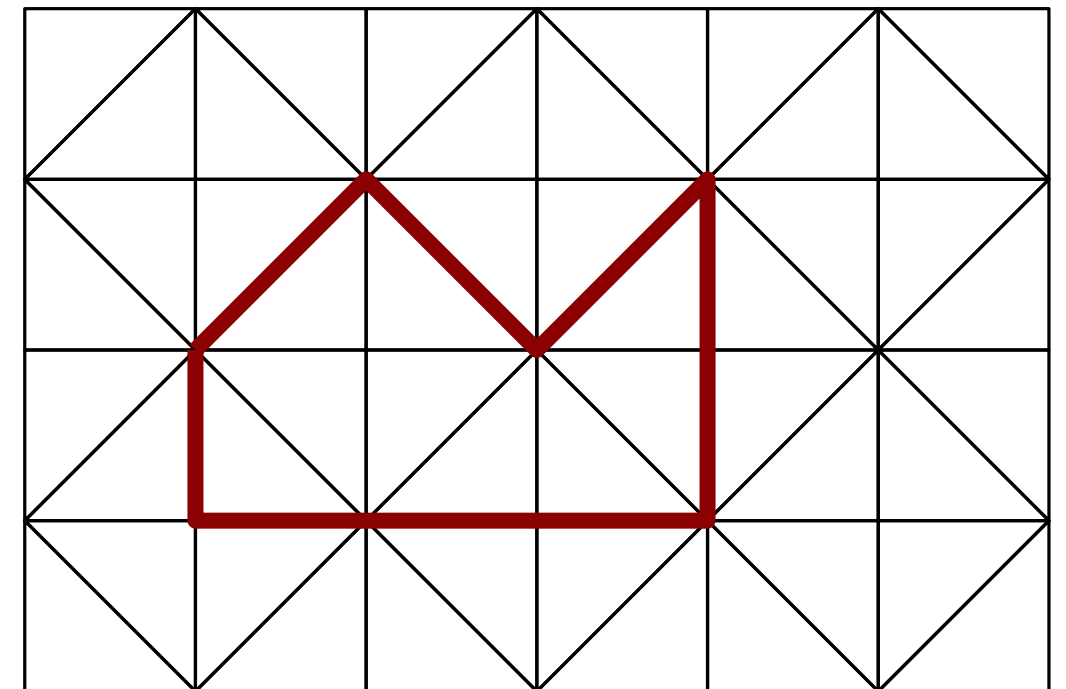


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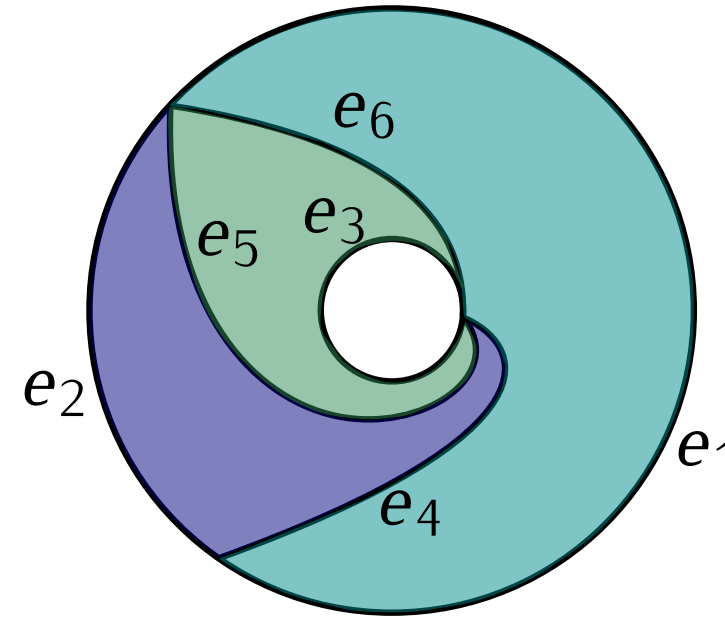
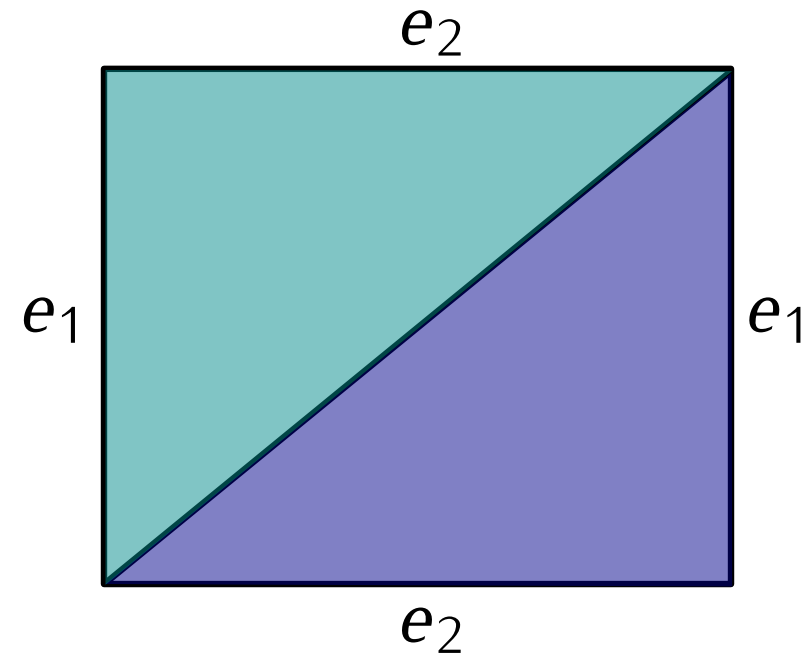


- A curve in a triangulation can be either
 - × **edged**: only lies on edges, is given by the list of edges where the curve lies

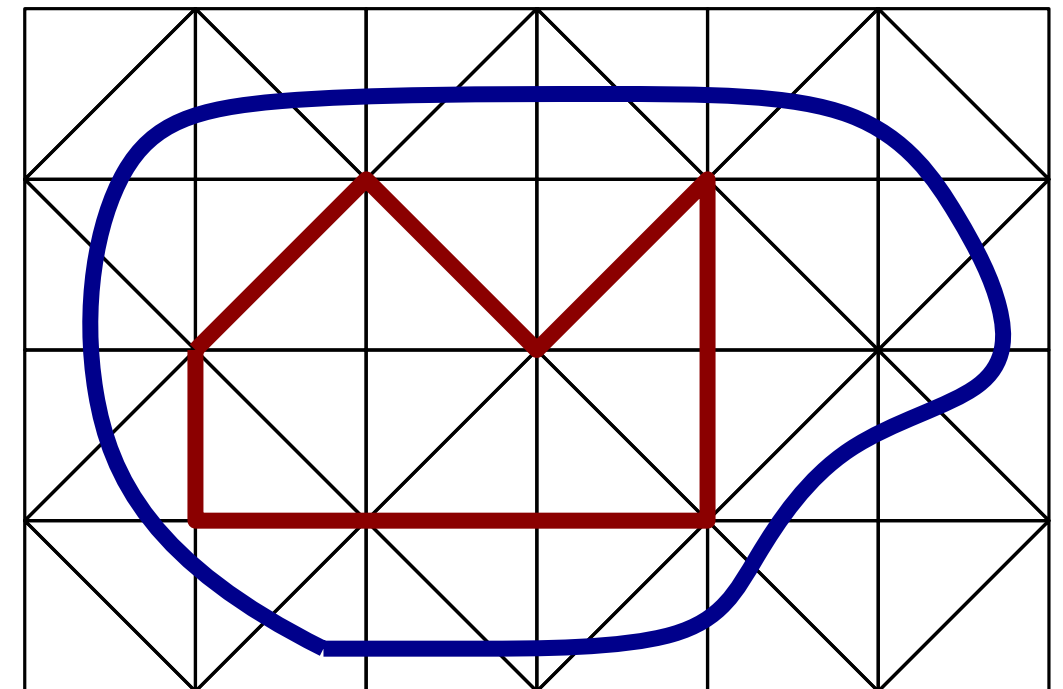


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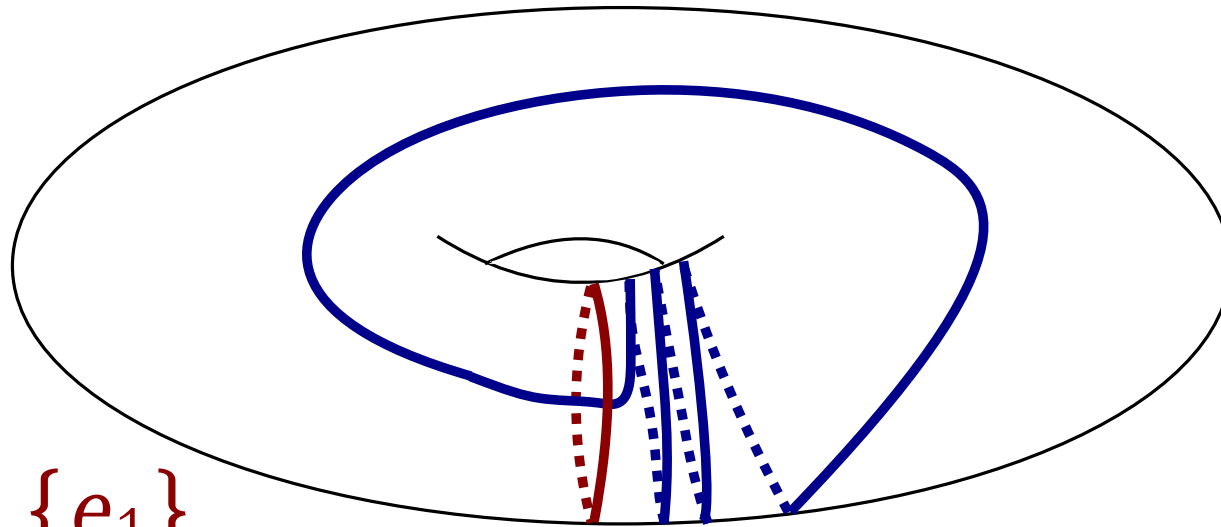


- A curve in a triangulation can be either
 - × **edged**: only lies on edges, is given by the list of edges where the curve lies
 - × **normal**: transverse to the edges, always enters and leaves a triangle through different edges, given by a vector with the number of times it intersects each edge



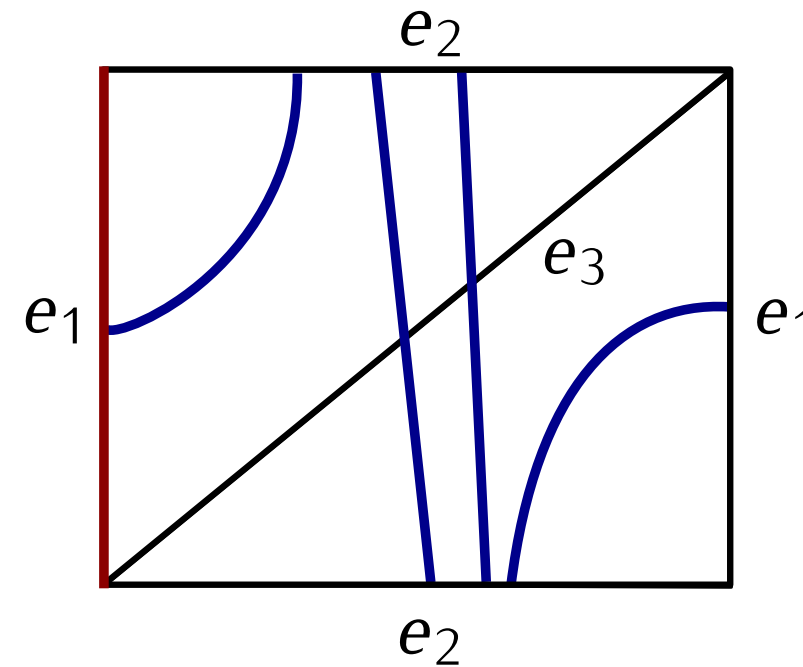
A data structure for Heegaard diagrams

- A Heegaard diagram $(\Sigma_g, \alpha, \beta)$ will be given as a compressed data structure $(T, E_T(\alpha), N_T(\beta))$



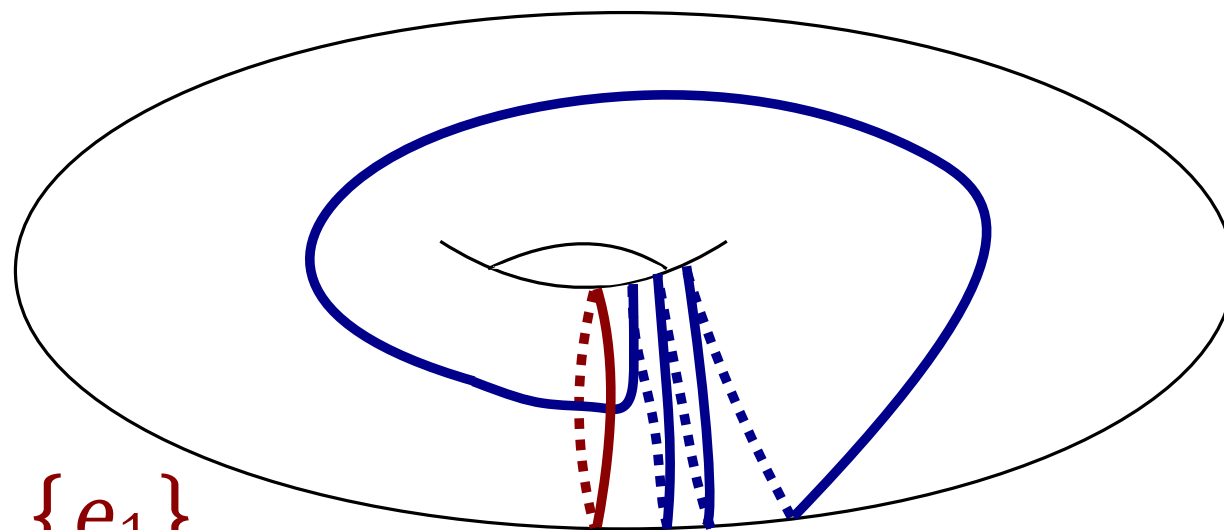
$$E_T(\alpha) = \{e_1\}$$

$$N_T(\beta) = \langle 1, 2, 2 \rangle$$



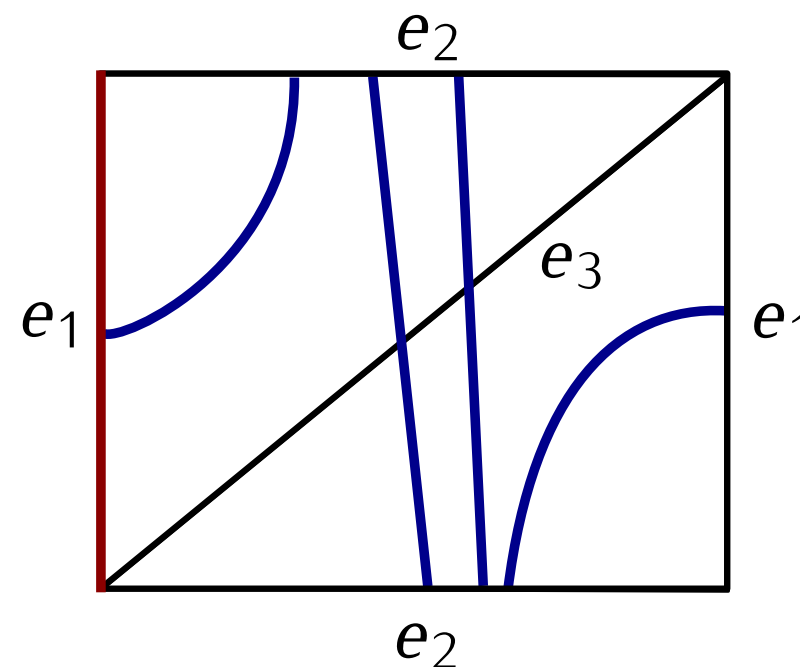
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$$E_T(\alpha) = \{e_1\}$$

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- If $\beta = \{b_i\}_{i=1}^g$, then the complexity of the splitting is the **binary** of the intersections with each edge

$$||N_T(\beta)|| = \sum_{i=1}^g \sum_e \log_2(|e \cap b_i|)$$

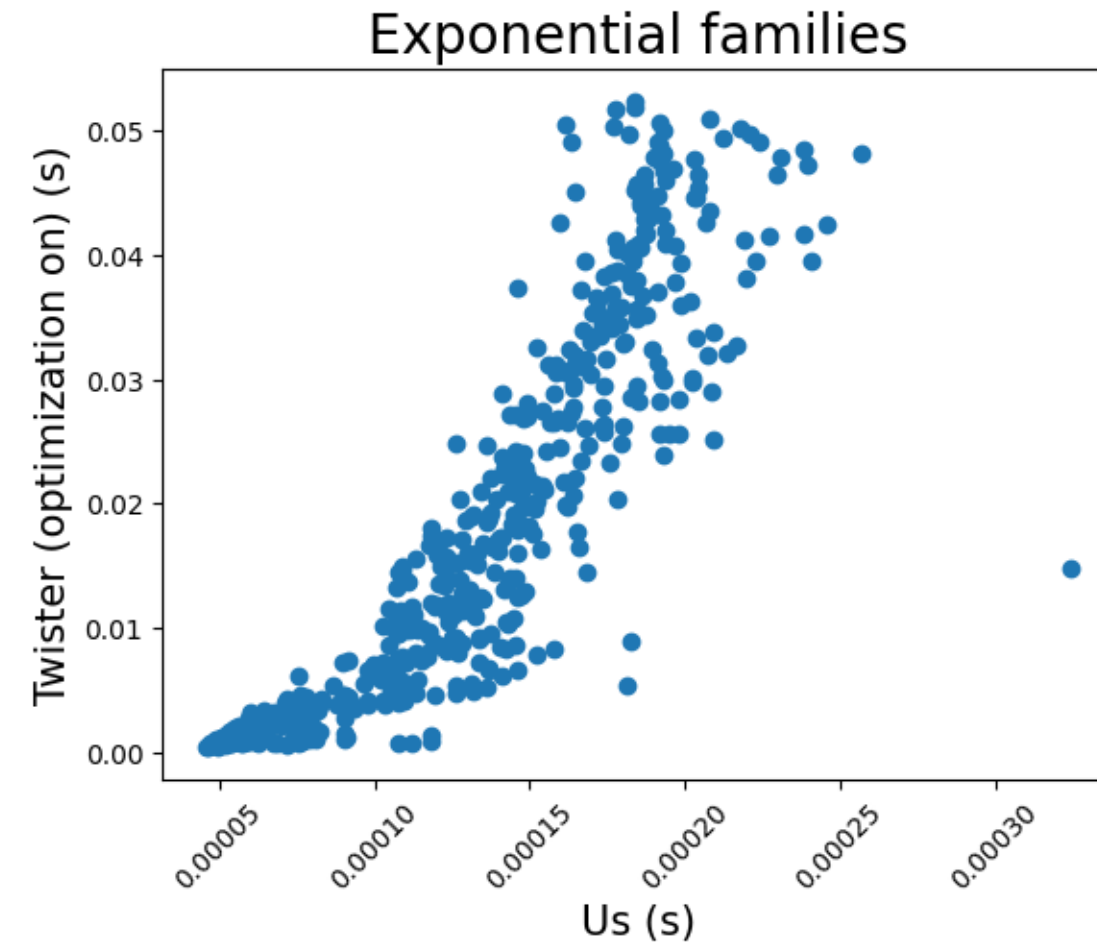
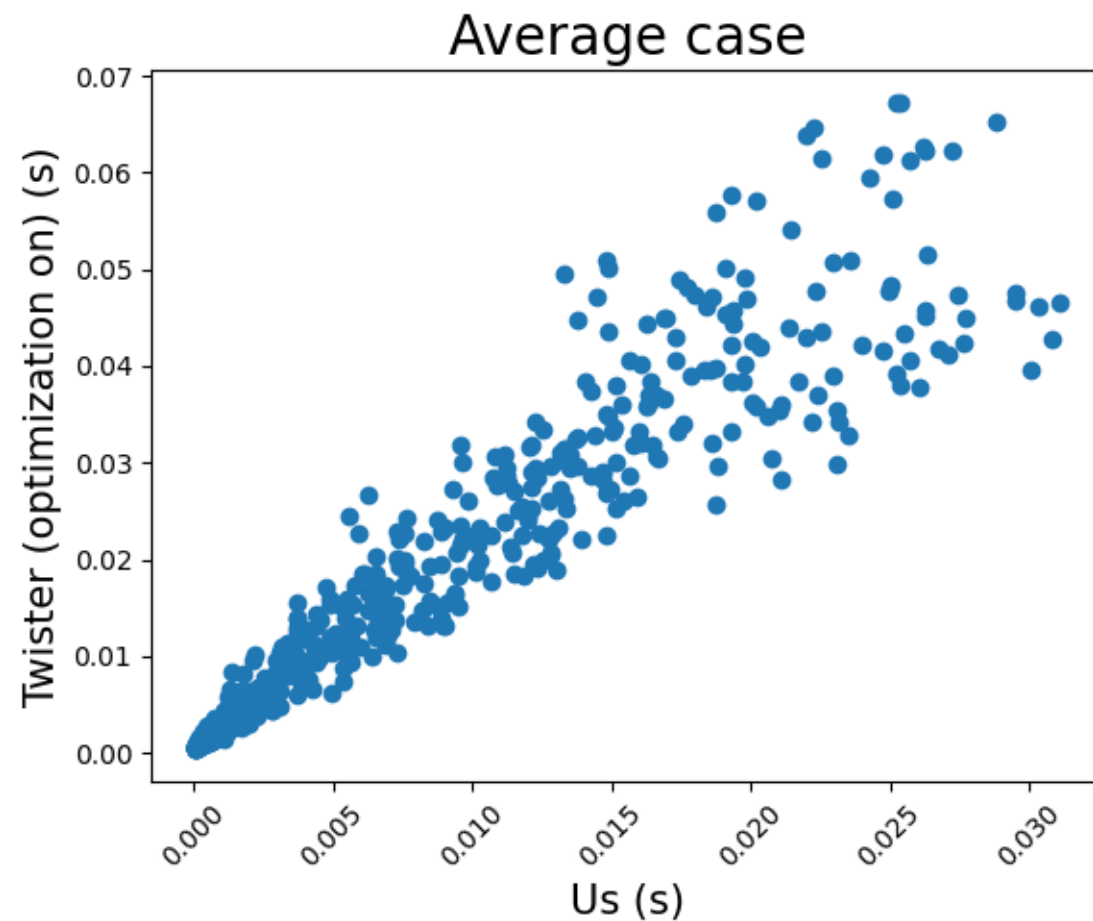
exponential many intersections of α and β

Some algorithms

Algorithm name	Description
CHECK DIAGRAM	Decide if curves define a diagram
GET EFFICIENT POSITION	Isotope α and β to efficient position
CHECK ISOTOPY	Check if two diagrams are isotopic to each other
DO A PACHNER MOVE	Do a Pachner move in the surface triangulation
DO A DISK SLIDE	Apply move to change systems
DETECT REDUCTION	Detect diagram reduction
DO STABILIZATION	Find a Heegaard diagram with larger genus
DO DESTABILIZATION	Find a Heegaard diagram with smaller genus
DO DEHN TWIST	Change the gluing map by a Dehn twist
COMPUTE π_1	Compute fundamental group
COMPUTE H_1	Compute first homology group

Some experiments

- We illustrate the power of FHDPy by computing H_1 for lens spaces ($g = 1$)



Some of the families constructed can only be triangulated with exponentially many tetrahedra, but have $||N_T(\beta)||$ linear on the indices

Thank you!